

ROSIA MONTANA, Romania

WMO: 151840

Lat: 46.3175N

Lon: 23.1392E

Elev: 3928

StdP: 12.73

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.0	(c) 10.6	(d) -3.0	(e) 5.4	(f) 16.4	(g) 2.0	(h) 7.1	(i) 17.7	(j) 27.0	(k) 28.9	(l) 23.0	(m) 27.5	(n) 7.2	(o) 40	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.6	(c) 77.1	(d) 63.0	(e) 74.3	(f) 61.9	(g) 71.9	(h) 60.8	(i) 65.0	(j) 73.8	(k) 63.5	(l) 71.8	(m) 62.2	(n) 70.1	(o) 6.2	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	96.1	68.8	60.4	90.9	67.7	59.0	86.4	66.5	32.3	74.3	31.0	71.8	30.0	70.4	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 21.9	(b) 18.4	(c) 15.9	(d) 15.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	
				(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	25.0	29.2	33.8	43.1	52.0	58.9	60.8	63.1	53.2	46.1	37.8	28.4
	DBStd	15.21	8.54	7.85	8.07	10.33	6.79	6.56	6.13	6.81	8.70	8.99	9.27	8.02
	HDD50	3479	775	583	504	250	58	8	3	2	68	186	373	669
	HDD65	7643	1240	1003	967	656	403	199	158	118	364	585	816	1134
	CDD50	1425	0	0	3	45	121	275	336	408	164	66	7	0
	CDD65	113	0	0	0	0	1	16	27	59	10	0	0	0
	CDH74	284	0	0	0	0	0	23	63	172	26	0	0	0
	CDH80	24	0	0	0	0	0	1	1	20	2	0	0	0
Wind	WSAvg	6.6	7.1	7.7	7.4	7.2	6.3	5.6	5.5	5.2	6.3	6.1	7.1	7.5
Precipitation	PrecAvg	31.9	1.7	1.5	1.9	2.7	3.4	4.0	4.2	3.0	2.8	2.4	2.2	2.4
	PrecMax	41.1	3.5	4.1	5.2	5.1	6.1	7.1	7.0	7.5	6.2	5.6	3.9	5.4
	PrecMin	22.1	0.5	0.3	0.5	0.4	1.9	1.3	1.2	0.7	0.4	0.1	0.0	0.2
	PrecStd	5.4	0.8	0.9	1.3	1.1	1.2	1.5	1.6	1.3	1.5	1.5	0.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	49.1	58.3	67.0	71.0	76.8	78.3	82.3	77.9	68.3	58.4	49.2
		MCWB	35.7	42.2	45.3	52.7	56.8	65.5	63.5	65.0	60.7	52.7	45.0	39.9
	2%	DB	40.9	45.1	52.5	63.5	67.7	73.6	75.5	78.3	71.8	64.3	54.6	44.9
		MCWB	35.2	39.5	42.8	50.6	55.8	62.8	62.9	62.6	58.8	52.6	45.2	39.2
	5%	DB	38.8	41.7	48.5	60.0	64.7	71.4	72.7	75.1	67.7	61.1	52.3	41.6
		MCWB	34.5	37.1	40.5	49.1	54.6	61.4	61.7	62.0	57.0	50.6	43.7	37.4
	10%	DB	36.6	39.3	44.7	56.5	62.0	68.7	69.8	72.3	64.7	58.1	49.8	39.0
		MCWB	33.1	35.8	38.5	47.2	53.4	60.3	60.5	61.1	55.7	50.1	43.0	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	45.0	47.2	54.5	58.9	67.3	65.7	66.6	62.0	56.3	49.3	42.9
		MCDB	42.6	47.7	56.8	65.2	67.2	75.1	73.7	77.5	73.8	61.9	54.7	45.6
	2%	WB	38.1	40.5	44.0	52.2	57.0	64.8	64.2	65.1	60.0	54.0	47.5	40.1
		MCDB	39.8	43.5	51.2	61.7	64.5	71.9	72.7	75.1	70.3	60.7	52.4	43.7
	5%	WB	36.3	38.3	41.4	50.2	55.6	62.7	62.9	63.4	58.1	52.4	45.9	38.2
		MCDB	37.8	40.9	48.0	58.8	63.2	70.0	70.7	72.5	66.6	59.8	50.5	41.0
	10%	WB	33.5	36.5	39.1	48.1	54.0	61.0	61.7	62.0	56.5	50.7	44.3	36.3
		MCDB	35.5	38.8	43.9	55.5	60.9	67.9	68.9	70.6	64.2	57.4	48.2	39.1
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	11.4	13.1	12.3	12.7	13.0	12.6	11.0	10.3	8.2	7.9
		MCDBR	8.5	10.6	15.4	15.7	15.3	15.4	14.7	14.5	13.5	12.7	10.2	9.4
		MCWBR	6.0	7.2	9.7	8.6	9.0	8.9	8.3	7.8	7.3	7.2	6.5	6.3
	5% WB	MCDBR	7.7	9.9	14.6	14.9	14.2	15.2	13.5	14.0	12.7	12.2	9.4	8.9
		MCWBR	6.3	7.4	9.5	8.6	8.9	9.2	8.5	8.0	7.4	7.5	6.3	6.8
Clear-Sky Solar Irradiance	taub	0.283	0.309	0.348	0.395	0.383	0.377	0.393	0.399	0.364	0.333	0.309	0.285	
	taud	2.379	2.331	2.294	2.193	2.279	2.320	2.276	2.258	2.336	2.420	2.440	2.416	
	Ebn at Noon	259	273	277	272	280	282	275	268	268	261	247	246	
	Edh at Noon	23	29	36	43	41	39	41	40	34	27	22	20	
All-Sky Solar Radiation	RadAvg	389	641	1003	1375	1653	1833	1796	1656	1178	812	469	310	
	RadStd	53	83	145	143	157	159	158	160	131	96	63	41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	+1.28	N/A	N/A	+1.80	N/A	N/A	-225	-340	+218	+108

Nomenclature: See separate page